

Mental health: data explainer (internal use only)

Allegra Saggese for Professors Galina Hale, Hanna Kim

June 10, 2026

1. Data sources

Section content

1. Data sources
2. Key files (csv)
3. Script pipeline overview
4. Merging into the panel
5. Review of panel dataset
6. Appendix

June 10, 2026

Allegra Saggese for Professors Galina Hale, Hanna Kim

1 / 58

Mental health: data explainer (internal use only)

Summary of data sources (1/2)

- **USDA NASS agriculture data:** Count of county-level CAFO operations classified as Small/Medium/Large by animal (hogs, chickens, cattle). Census years only: 2002, 2007, 2012, 2017. Pulled via USDA Quick Stats API.
- **USDA FSIS data:** Count of federally-inspected slaughterhouses and meat processing plants by size bucket and animal type at county-level. Source: FOIA request. Geographic recovery via HUD ZIP-to-FIPS crosswalk.
- **CDC mortality data:** Deaths of despair (drug overdose + suicide + alcohol) at county-level from CDC WONDER, annual. Crude rates suppressed for counties with < 10 deaths.

Summary of data sources (2/2)

- **County Health Rankings (CHR) data:** Self-reported survey measures of mental health and health behaviors at county-level. Annual coverage from ~2004. Variable selection based on cross-year availability.
- **Crime data:** FBI UCR incident-level arrests collapsed to FIPS-year. Violent crime subset selected by research team.
- **Population and geographic data:** US Census Bureau county population (1990–2024). NCHS urban-rural classification (6-level) used to define the main analytical sample.

2. Key files (csv)

Section content

1. Data sources
2. Key files (csv)
3. Script pipeline overview
4. Merging into the panel
5. Review of panel dataset
6. Appendix

June 10, 2026

Allegra Saggese for Professors Galina Hale, Hanna Kim

3 / 58

Mental health: data explainer (internal use only)

File locations: Shared directory structure

All data lives in the shared `~/Dropbox/Mental/` folder. **Do not move or rename top-level folders, as the scripts are set to run from the dropbox structure.**

Starting with the raw data files: **Raw inputs** (`Data/raw/`)

- `cdc/` — CDC WONDER mortality CSVs cleaning script
- `crime/total-v1/` — FBI UCR (pre-assembled) annual CSVs cleaning script
- `fips/` — FIPS crosswalk (county-year key)
- `mental/` — County Health Rankings CSVs cleaning script
- `nchs/` — `NCHSurb-rural-codes.csv` details
- `population/` — Census population CSVs details
- `usda/` — USDA agriculture census `.dta` fallback files cleaning script
- `fsis/` — FSIS (FOIA-requested) inspection CSVs cleaning script

June 10, 2020

Mental health: data explainer (internal use only)

File locations: Shared directory structure

Clean outputs are the cleaned versions of each input source. If source was stored across multiple files (csvs, .dta files), the clean output contains the consolidated year-county version of the data.

Next we have interim, but organized **clean outputs** (Data/clean/)

- YYYY-MM-DD_population_full.csv
- YYYY-MM-DD_cafo_ops_by_size_compact.csv
- YYYY-MM-DD_mh_mortality_fips_yr.csv
- YYYY-MM-DD_mentalhealthrank_full.csv
- YYYY-MM-DD_cdc_county_year_deaths_of_despair.csv
- YYYY-MM-DD_crime_fips_level_final.csv
- YYYY-MM-DD-rural-key.csv

Data/clean/ files: unit of observation & contents (1/4)

File	Unit of obs.	Years	Descriptor + key variables
population_full.csv	county-year	1990–2024	Annual Census population, 4 vintages spliced & FIPS-expanded. Raw: population, state/division/region. 100% coverage — panel denominator.
fips_full.csv	county-year (ref.)	1990–2024	FIPS↔county-name crosswalk, expanded to annual rows per census-decade vintage. Raw reference table for key normalization (not merged into panel).

Data/clean/ files: unit of observation & contents (2/4)

File	Unit of obs.	Years	Descriptor + key variables
cafo_ops_by_size_compact.csv	county × year × commodity (5 types)	2002, 2007, 2012, 2017, 2022	USDA NASS operation counts by size. Raw: small, medium, large, bin_sum, total_heads. Derived: any_large_cafo, any_medium_or_large_cafo (presence flags).
cdc_county_year_deaths_of_despair.csv	county-year	2000–2020	CDC WONDER deaths of despair (drug overdose + suicide + alcohol). Raw: deaths, population, crude_rate, crude_rate_raw, is_unreliable, pct_of_total_deaths. Crude rate suppressed < 10 deaths.

June 10, 2026

Mental health: data explainer (internal use only)

Data/clean/ files: unit of observation & contents (3/4)

File	Unit of obs.	Years	Descriptor + key variables
crime_fips_level_final.csv	county-year	1991–2021	FBI UCR violent-crime incident counts, collapsed from precinct level. Raw: 12 incident-type counts (aggravated_assault, rape, simple_assault, etc.). Derived: total_incidents (row sum).
mentalhealthrank_full.csv	county-year	2010–2023	County Health Rankings survey, ~50 measures. All raw as reported by CHR: *_raw_value, *_numerator, *_denominator, *_ci_high/low. (Per-100k & count-imputed versions built later, in the panel.)

June 10, 2026

Mental health: data explainer (internal use only)

Data/clean/ files: unit of observation & contents (4/4)

File	Unit of obs.	Years	Descriptor + key variables
...-rural-key.csv	county-year	2000–2023	NCHS urban-rural codes, forward-filled from 4 base years (1990/2006/2013/2023). Raw: <code>nchs_code</code> , <code>nchs_label</code> , <code>cbsa_pop</code> , <code>county_pop</code> . Derived: sample filter → becomes rural in the panel.
..fsis_county_year...summary...csv	county-year	2017–2026	FSIS-regulated establishment counts by type & size bucket; FIPS recovered via HUD ZIP crosswalk. All raw counts: <code>n_unique_establishments</code> , <code>n_meat_slaughter_establishments</code> , <code>n_size_bucket_1-5</code> , etc.

June 10, 2026

Mental health: data explainer (internal use only)

File locations: Complete panel

Merged panel: (Data/merged/): YYYY-MM-DD_full_merged.csv — the main analysis file.
Always use the most recent date prefix.

script1c-merge-dataclean.py is the only script that writes
YYYY-MM-DD_full_merged.csv, and that file is the *one and only analysis panel*.
Export on line 794-797. Everything in the set of script2* reads this file and produces
figures or quality assurance outputs. There is no modification or re-export of the panel.

script pipeline details

File locations: Where (and what) are the key files?

Use Case	File (assume YYYY-MM-DD prefix)	Location
Full panel (all years)	<code>_full_merged.csv</code>	Data/merged/
2005–2010 slice	<code>_full_merged_2005_2010.csv</code>	Data/merged/
2010–2020 slice	<code>_full_merged_2010_2020.csv</code>	Data/merged/
Census years only	<code>_full_merged_census_years.csv</code>	Data/merged/
CAFO only	<code>_cafo_ops_by_size_compact.csv</code>	Data/clean/
FSIS only	<code>fsis_county_year_..._summary.csv</code>	Data/clean/

Note: All files are date-stamped. Function `latest_files_by_descriptor()` in `functions.py` auto-selects most recent version by filename descriptor string — no manual tracking required for script re-runs.

June 10, 2020

Mental health: data explainer (internal use only)

Merged panel (`panel.csv`): unit of observation & overview

Unit of observation: county-year (`fips`, `year`) — 162 columns, 65,828 rows, 2,879 unique rural counties.

Years: source-dependent, ~1990–2023 overall; recommended window **2010–2020** (see coverage table later in deck).

Slices (`panel_05_10`, `panel_10_20`, `panel_census_years`) share this schema. `panel_fsis.csv` (155 cols, 2017+ only) additionally includes the FSIS establishment columns.

Reading the inventory (next 3 slides): “Raw” = passthrough from a Data/clean/source file (slides above). “Derived/transformed” = computed in `script1b-generate-panel.py` (rates, ratios, flags, aggregates). No raw values are

overwritten — derived columns are always additional columns. Metadata: data explainer (internal use only)
June 10, 2026
Allegra Saggese for Professors Galina Hale, Hanna Kim 12 / 58

Merged panel (panel.csv): variable inventory (1/3)

Category	Raw columns (examples)	Derived/transformed columns (examples)
CAFO ops	cafo_{animal}_{size} (15 cols), {small/medium/large/ bin_sum/total_heads}_cafo (summed across the 5 commodities)	cafo_total_ops_all_animals, cafo_total_ops_chickens, cafo_beef_total, cafo_dairy_total, any_large_cafo, any_medium_or_large_cafo
Crime (UCR)	12 incident counts + total_incidents (_crime suffix)	*_per100k (13 cols, standardized to per-population)

Merged panel (panel.csv): variable inventory (2/3)

Category	Raw columns (examples)	Derived/transformed columns (examples)
CDC mortality	deaths_despair, crude_rate_despair, cdc_deaths, cdc_population, cdc_crude_rate	crude_rate_from_census_pop, deaths_per_10k_census_pop, cdc_in_query, deaths_is_zero, crude_rate_recalc_* (sense-check)
CHR survey	*_raw_value (e.g. poor_mental_health_days, adult_obesity, frequent_mental_distress)	*_per100k (27 cols, standardized incidence), *_count_imputed (6 cols, rate $\times$ population)

Merged panel (panel.csv): variable inventory (3/3)

Category	Raw columns (examples)	Derived/transformed columns (examples)
Population & geography	population, state, state_code, state_fips, state_abbrev, division, region	census_population (alias used in CDC sense-check block)
Sample filter	—	rural (always 1; from NCHS non_large_metro restriction)

Disaggregated panel

We want to have an option to create a disaggregated (by demographic) panel that offers further classification by **sex, race, hispanic/non-hispanic, age**. What we actually can do:

- Age data is only available for arrest records. We cannot use this for the rest of the data.
- Sex available for all data
- Race available for all data, although CHR does not allow for intersection identification (black + hispanic)

What we have: `script0c` allows for sex + race disaggregated analysis, use `RUN_CDC_DISAGG_EXTENSION` by unhashing in the code.

3. Script pipeline overview

Section content

1. Data sources
2. Key files (csv)
3. Script pipeline overview
4. Merging into the panel
5. Review of panel dataset
6. Appendix

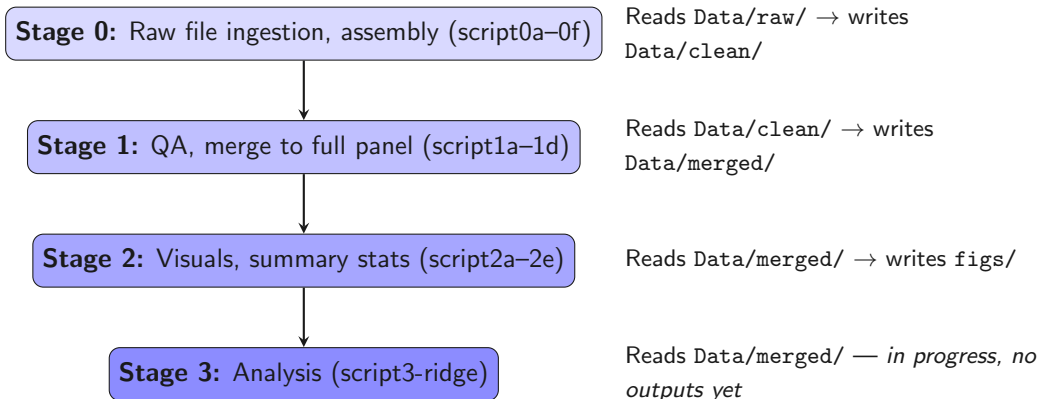
June 10, 2026

Allegra Saggese for Professors Galina Hale, Hanna Kim

16 / 58

Mental health: data explainer (internal use only)

Script pipeline: execution order for cleaning-analysis



Script pipeline: helpers

The github repo, `mentalhealth` contains all of the scripts directly (no scripts execute from subfolders), that are needed to run project start to finish. Each script is clearly labeled. Two **key utility scripts** exist with helper functions used in multiple scripts.

These include:

- 1 `packages.py` sets all packages that are necessary for analysis, has all shared import/export locations and other Dropbox path roots (interim-data, QA outputs, allegra-dropbox-copy for manual inspections)
- 2 `functions.py` provides FIPS normalization, CSV fallback readers, API wrappers, and the `latest_files_by_descriptor()` selector used in Stage 1.

script0a: Population + FIPS Panel

Purpose: Build a complete annual FIPS-year population panel from 1990–2024 using four census vintages. Census vintages come in 10 year intervals (1990-2000, 2000-2010, etc.) So we use three 10-year cycles and then 2020-25 data. FIPS are provided on a census year basis, same 10-year interval update.

Inputs: Data/raw/population/*.csv (4 Census files) and Data/raw/fips/foruse_*.txt (pipe/comma/whitespace delimited crosswalks)

Output: Data/clean/YYYY-MM-DD_population_full.csv — **100% county-year coverage**, used as second-best source population denominator throughout the project.

Key cleaning steps:

1. Reads each Census file by fragment name match (`cc-est2024`, `cc-est00int-tot`, etc.)
— **if filenames change, these fragment strings must be updated**
2. Identifies FIPS crosswalk delimiters via `sniff_delimiter()` and reads accordingly (pipe / comma / fixed-width)
3. Expands each decade's crosswalk to annual rows via `expand_years_cross()`
4. **Hardcoded manual fix:** FIPS 12025 → 12086 (Miami-Dade) and two counties added for 1990 (Yellowstone NP 30113, South Boston 51780)
5. Drops state-level summary rows where `county == state`
6. Normalizes DC to 11001; drops non-US state codes; deduplicates on (`fips`, `year`) keeping last (QA shows no dupes exist, one FIPS per year per county name)

script0b: USDA NASS Agriculture (CAFO)

Purpose: Pull all agriculture operation counts by animal and size class from USDA NASS Quick Stats API (census years: 2002, 2007, 2012, 2017). Agriculture operations include any farming (non-processing or slaughter) establishment, and is classified by type of animal and purpose of breeding (i.e. cattle for milk vs. beef).

Inputs: Census API. Where API doesn't work, we use pre-downloaded .dta files as previously saved in Data/raw/usda/*.dta

Outputs: *_cafo_ops_by_size_compact.csv (pivot: county $\times$ year $\times$ commodity $\times$ small/medium/large), establishment counts at county-level are intergrated into merged_all.csv

Key assumptions, data cleaning step 1:

- **API key:** `os.environ["USDA_NASS_API_KEY"] = "30643212-..."` — **must be changed per user**
- **API fallback:** If API fails, reads `.dta` files from `Data/raw/usda/`
- **backfill for non-census years.** Using the previous four years, we use non-transformed backfilled data from most recent census year to fill in for annual basis.
- Filters to: CENSUS source, COUNTY level, animals = {CATTLE, CHICKENS, HOGS}, unit = {HEAD, OPERATIONS}
- Splits queries $> 50,000$ rows by domain to avoid API limits (rate sleep: 0.4s)

Data creation: Size classification thresholds for CAFOs: Using the USDA CAFO definition (on animal head count parameter only), applied per inventory bin code. Note the bins are a translation of *ranges* of number of animal heads.

Animal	Medium $\geq$ bin	Large $\geq$ bin
Broiler chickens	3	5
Layer chickens	7	9
Cattle / Hogs	6	7

script0c: Health Data (CHR + CDC Mortality)

Purpose: Clean County Health Rankings (CHR) mental health surveys and CDC deaths-of-despair county files; merge into a single health panel.

Inputs: CHR mental health survey data found in `Data_mental_analytic*.csv`. These are merged. Then CDC data exported from WONDER as saved in `Data_cdc_cty-level-deaths-of-despair*.csv`.

Outputs: Clean mental health survey data in `*_mentalhealthrank_full.csv`, `*_cdc_county_year-deaths-of-despair.csv`,

depreciated output: `*_mh_mortality_fips_yr.csv`

CHR cleaning (Pass 1):

- Year inferred from filename via regex from the 4-digit year (on all files)
- Column names normalized: lowercased, special characters stripped, label drift harmonized (e.g. limited access to healthy foods → access to healthy foods)
- Duplicate columns resolved by keeping the most complete version (highest fill rate)
- **Variable selection thresholds:** include if present in ≥ 11 annual files (majority rule) OR ≥ 8 files with $\geq 80\%$ average fill
 - a Threshold can be changed in the script
 - b Conservative threshold chosen first

CDC cleaning (Pass 2):

- Reads files matching `cty-level-deaths-of-despair-YYYY.csv` from `Data/raw/cdc/`
- Parses: deaths, population, crude rate, `is_unreliable` flag (from “unreliable” / “suppressed” in notes)
- **Suppression caveat:** crude rate is missing for ~85–93% of rural county-years (CDC suppresses when deaths < 10). Raw death count is always retained.
- Using the population data from `script1a-pop-fips.py` we can calculate the incident rate of deaths
- Note: missing values from CDC are assumed to be missing data, and are kept separate from values classified as zero. QA check performed to ensure mislabeling does not occur.

Merge: Outer join of CHR + CDC on (`fips`, `year`); validated as 1:1.

June 10, 2026

Mental health: data explainer (internal use only)

script0d: Crime Data (FBI UCR)

Purpose: Stack annual FBI UCR incident files as provided from 2023 analysis (precinct → county aggregation). Then we select violent crime columns (researcher discretion), and collapse the number of incidence to FIPS-year counts.

Inputs: Loads all .csv from Data/raw/crime/total-v1/ (~7 GB combined in memory — beware on low-RAM machines)

Outputs: *_crime_fips_level_final.csv (main), *_crime_demog_final.csv (with demographics)

Key steps:

1. **Crime types** selected by research team: aggravated assault, DUI, fondling, incest, intimidation, kidnapping, rape, sexual assault, simple assault, statutory rape, human trafficking (commercial sex + involuntary servitude)
2. Column normalization: unicode dash normalization, whitespace collapse, then fuzzy matching for human trafficking columns (some encoding issues required manual rename map)
3. Collapses first to (fips, year, sex, race, ethnicity) keys, then **further to** (fips, year) **only** for the main panel
4. **Optional disaggregation extension** (off by default): set env var `RUN_CRIME_DISAGG_EXTENSION=1` to produce sex/age/race breakdowns in `interim-data/crime-disagg-extension/`

script0e: NCHS Urban-Rural Classification

Purpose: Build an annual rural/urban classification panel from the NCHS 6-level code scheme, used to define the analytical sample. Panel restricted to years ≥ 2000 .

Inputs: Using `Data/raw/nchs/NCHSurb-rural-codes.csv`, containing year x county codes for given base years (1990, 2006, 2013, 2023).

Output: `YYYY-MM-DD-rural-key.csv` — the filter applied in `script1c` to drop all non-rural counties before creating `YYYY-MM-DD-final-merge.csv`

NCHS 6-level scheme:

Code	Label	non_large_metro
1	Large central metro	0 (excluded)
2	Large fringe metro	0 (excluded)
3	Medium metro	1 (included)
4	Small metro	1 (included)
5	Micropolitan	1 (included)
6	Noncore (rural)	1 (included)

Data manipulation for this dataset is simple. No changes to data structure are needed. Backfilling for missing years (non-recorded years) are needed.

Annual expansion logic: NCHS provides codes at four base years. Each base year is forward-filled to cover the next period:

- 1990 code → 1990–2005;
- 2006 code → 2006–2012;
- 2013 code → 2013–2022;
- 2023 code → 2023

script0f: FSIS processing establishments pipeline

Purpose: Multi-step pipeline to clean FOIA-requested FSIS inspection data, recover missing FIPS codes for establishments in the FSIS data via the HUD ZIP-to-FIPS crosswalk, and create a county x year dataframe of inspected facilities.

Requirements: HUD_API_TOKEN env var required (unless `--hud-strategy skip`).

Output:

*_fsis_county_year_fips_est_size_type_summary_hudbulk_manualzip.csv, and inclusion in FSIS facilities into final merged panel

script0f: FSIS processing establishments pipeline

Pipeline sub-scripts (run in order):

1. `script0e-fsis-slaughterhouses.py` — extract raw FSIS inspection records
2. `script0h-fsis-establishment-size-panel.py` — build establishment × size panel
3. **HUD fill** (choose strategy via `--hud-strategy`):
 - ▶ bulk (default): ZIP-to-FIPS crosswalk for all establishments with city/state/ZIP
 - ▶ targeted: only missing establishments
 - ▶ skip: no HUD fill
4. `script0l` — apply manual ZIP template (if present in `qa-fsis/`)
5. `script0m` — deterministic FIPS completion
6. `script0n` — apply manual county labels (if present)

4. Merging into the panel

Section content

1. Data sources
2. Key files (csv)
3. Script pipeline overview
4. Merging into the panel
5. Review of panel dataset
6. Appendix

June 10, 2026

Mental health: data explainer (internal use only)

script1c: The master merge file

Purpose: Combine all clean source files into one county-year panel, restricted to rural counties, with all iterated or re-calculated columns for analysis.

Step-by-step logic:

1. **Auto-selects latest files** from Data/clean/ by descriptor string (date prefix stripped)
2. **Builds rural allowed-keys:** reads *-rural-key.csv, keeps only (fips, year) where `non_large_metro == 1`
3. **Base panel:** CAFO compact file reduced to one row per (fips, year) — numeric columns summed (`min_count=1` to preserve NaN for all-missing groups); text columns take first observed value
4. **CAFO animal × size block:** pivot adds columns
`cafo_{cattle/hogs/chickens}_{small/medium/large},`
`cafo_total_ops_all_animals, cafo_total_ops_chickens`

June 10, 2020

Mental health: data explainer (internal use only)

Cont'd **step-by-step logic**:

5 **Left-joins** CDC mortality, crime, FSIS, CHR mental health, and population onto the rural CAFO base frame.

► *Note: Final QA is needed to ensure rural county dataframe is used as base, not CAFO dataframe, so we don't exclude zero-presence of ag activity counties.*

6 **Derived crude rate:** $\text{crude_rate_from_census_pop} = \text{deaths} / \text{census population} \times 100,000$ — computed only where deaths > 0 and population > 0. This is done for *crude_rate* mortality data flagged as unreliable in source file.

7 **Missingness flags:** `cdc_in_query`, where `deaths_is_zero` flag captures missing CDC data. CDC returned 0 deaths — ambiguous, rate set to NaN.

script1c: Assumptions & caveats

- **Rural filter is the primary sample restriction.** Large central and fringe metro counties are dropped entirely.
- **Column naming convention:** All merged columns are suffixed with the descriptor of their source file (e.g. `population_population_full`). Do not rename these manually.
- **Summing with `min_count=1`:** When collapsing duplicates, numeric columns are summed but an all-missing group is preserved as NaN rather than coerced to 0.
- **Text column resolution:** String columns from multi-row sources take the *first* non-null observed value after the rural filter. This is appropriate for geographic labels, not used for time-varying text fields.

script1e: Derived variables (CHR counts + ratio flags)

Purpose: Two-pass augmentation of the merged panel using County Health Rankings (CHR) data. Recall the CHR data is survey data at the county level. The data provides raw data, crude rates of incidence, numerator, denominator and confidence intervals (CIs). CIs cannot be used, but to calculate crude rates, we use population in the panel key for consistency in calculations.

Output: *_full_merged_derived.csv (original panel unchanged). Note, in the most recent version of the code, this has been dropped and now *_full_merged.csv contains the changes made in this script.

script1e: Derived variables (CHR counts + ratio flags)

Pass 1: Impute counts for rate-only CHR variables:

- Applies to 6 variables that have `raw_value` only (no numerator column): frequent mental distress, frequent physical distress, insufficient sleep, access to exercise, % American Indian/Alaskan Native, % non-Hispanic African American
- Formula: `count_imputed = (raw_value / 100) × census_population`
- Scale auto-detected: $\text{median} \leq 1 \Rightarrow$ proportion (multiply by 100); otherwise already a percentage
- **These are imputed, not measured counts.** Label: `*_count_imputed`

script1e: Derived variables (CHR counts + ratio flags)

Pass 2: Verify numerator/denominator consistency:

- For all CHR variables with both `numerator` and `denominator` columns, checks:
`numerator / denominator  $\approx$  raw_value`
- Tolerance: absolute diff > 0.05 (5 pp) OR relative diff > 10% $\Rightarrow$ flagged
- Flag column: `*_ratio_flag = 1`. Variables with > 5% rows flagged generate a warning.
- **No values are modified** — flags only. Inspect flagged variables before use in regression.

$\Rightarrow$ we find no issue with data assembly by CHR. Survey method does not yield obvious issues or column discrepancies.

Pass 3: Build standard incidence rates to per 100k rate:

- The incidence rates in the CHR data are reported with different denominators
- In `script1c` we align crime and health data on rates of incidence for comparison
- Incidence rate columns are pulled from CHR data, and depending on presentation, are converted. [See details in appendix](#)

script1d: CDC crude rate sense check (population)

Purpose: Validate that CDC-reported crude rates align with rates recomputed from merged deaths and census population. Flags systematic discrepancies.

Diagnostics produced:

- `crude_rate_recalc_census_pop` = $\text{deaths} / \text{census pop} \times 100,000$ (recomputed)
- `crude_rate_recalc_cdc_pop` = $\text{deaths} / \text{CDC pop} \times 100,000$ (internal check)
- `diff_census_minus_cdc`: difference between recomputed and CDC-reported rates
- `z_diff_from_zero_sd`: standardized distance of discrepancy from zero
- `population_pct_diff_census_vs_cdc`: % difference between Census and CDC population denominators

script1d: CDC crude rate sense check (population)

QA outputs (saved to Data/merged/figs/panel-sumstats-by-farms/):

- *_qa_cdc_cruderate_sensecheck_overall.csv — global summary statistics
- *_qa_cdc_cruderate_sensecheck_by_year.csv — year-by-year breakdown
- *_qa_cdc_cruderate_sensecheck_outliers_top1000.csv — largest discrepancies

Note there are **hardcoded paths in this script**: MERGED_DIR and OUT_QA_DIR are absolute paths — update before running on a new machine.

Population comparison demonstrated CDC, CHR and US census population data was not different enough to flag between sources.

5. Review of panel dataset

Section content

1. Data sources
2. Key files (csv)
3. Script pipeline overview
4. Merging into the panel
5. Review of panel dataset
6. Appendix

June 10, 2026

Allegra Saggese for Professors Galina Hale, Hanna Kim

42 / 58

Mental health: data explainer (internal use only)

Merged panel: what is it?

Recall our final **file**: `Data/merged/YYYY-MM-DD_full_merged.csv`

Unit of observation: county-year (fips, year)

Sample restriction: non-large-metro counties only (`non_large_metro == 1`) for 2,314 counties per year

Year range: ~2000–2023 (source-dependent). 2010–2020 offers most complete set of data from mental health survey, crime and agriculture data. See `YYYY-MM-DD_full_merged_2010–_2020.csv`

Panel structure: unbalanced — not all sources cover all years. In total: we have 65,828 rows and 2,879 unique (rural) counties.

Merged panel: what is it?

Column naming convention:

All non-key columns are suffixed with their source descriptor:

- `...cafo_ops_by_size_compact` — CAFO columns
- `...cdc_county_year_deaths_of_despair` — CDC mortality
- `...mental_health_rank_full` — CHR survey
- `...crime_fips_level_final` — UCR crime
- `...fsis_county_year...summary...` — FSIS plants
- `...population_full` — Census population

Key variables of interest(1/2)

Variable	Description	Source
cafo-{animal}-{size}	Count of small/medium/large CAFO ops by animal type (cattle, hogs, chickens)	USDA NASS census years
cafo_total_ops_all_animals	Total CAFO operations across all animal types	Derived in script1c
population_population_full	Annual county population (Census)	script0a — 100% coverage
deaths_cdc_...	Raw deaths-of-despair count	CDC WONDER
crude_rate_cdc_...	CDC-reported crude rate (suppressed < 10 deaths)	CDC WONDER

June 10, 2026

Mental health: data explainer (internal use only)

Key variables of interest (2/2)

<code>crude_rate_ from_census_pop</code>	Recomputed crude rate using Census pop denominator	Derived in script1c
<code>cdc_in_query</code>	1 if county-year appeared in CDC WONDER download	Derived flag
<code>deaths_is_zero</code>	1 if CDC returned Deaths=0 (ambiguous)	Derived flag
<code>poor_mental_health_ _days_raw_value_...</code>	Avg days/month of poor mental health (CHR)	County Health Rankings
<code>frequent_mental_ distress_raw_value_...</code>	% adults with frequent mental distress (CHR)	County Health Rankings
<code>n_unique_establishments_ _fsis_...</code>	Count of FSIS-regulated plants per county (2017 only)	USDA FSIS FOIA
<code>aggravated_assault</code>	UCR aggravated assault count	FBI UCR

June 10, 2026

Mental health: data explainer (internal use only)

Key variables of interest: coverage (1/2)

Source	Year range	Fill (% panel)	Notes
Census population	1990–2024	$\approx 100\%$	Used as denominator
CAFO operations	2002, 2007, 2012, 2017, 2022	$\approx 97\%$ (census yrs)	Census years only
CHR poor mental health days	2004–2022	$\approx 90\%$ (2010+)	Usable 2010 onward
CHR frequent mental distress	2016–2022	$\approx 90\%$ (2016+)	Not for pre-2016 panel
CHR excessive drinking	2011–2022	$\approx 80\%$ (2011+)	Full from 2016

June 10, 2026

Mental health: data explainer (internal use only)

Key variables of interest: coverage (2/2)

CDC deaths of despair (raw count)	1999–2020	Varies	Count retained always
CDC crude rate	1999–2020	< 15%	Suppressed < 10 deaths
FBI UCR crime	~2000–2022	Varies by year	Violent crime subset
FSIS establishments	2017 only	2017 cross-section	Cross-section use only

Recommended panel window for outcome analysis: 2010–2015. This balances CHR mental health coverage ($\approx 90\%$), CAFO census year availability (2012), and pre-FSIS comparability.

Known issues and how to handle them:

1. **CDC crude rate missingness ($\approx 85\text{--}93\%$):** This is by design — CDC suppresses for < 10 deaths. Use raw death counts or the recomputed `crude_rate_from_census_pop` for analysis. Flag `cdc_in_query = 0` means the county-year was absent from the CDC WONDER pull entirely.
2. **CAFO data at census years only (2002, 2007, 2012, 2017, 2022):**
Interpolating between census years requires justification in explanation.

Known issues and how to handle them:

1. **CHR** `frequent_mental_distress` **starts 2016**: Cannot be used in a pre-2016 panel. Use `poor_mental_health_days` (available from ~2004, reliable from 2010) as the primary mental health outcome for longer panels.
2. **CAFO per-capita transformation**: Raw CAFO operation counts are heavily right-skewed. For scatter plots and regressions, use $\log(\text{cafo_total_ops_all_animals} / \text{population} \times 10,000 + 1)$.

Time-dependent slices

File suffix	Year range	Recommended use
<code>_full_merged.csv</code>	All years (rural)	Starting point for all analysis
<code>_full_merged_2005_2010.csv</code>	2005–2010	Pre-recession window
<code>_full_merged_2010_2020.csv</code>	2010–2020	Recommended primary panel
<code>_full_merged_census_years.csv</code>	2002, 2005, 2007, 2012, 2017, 2022	CAFO-aligned cross-sections
<code>_full_merged_derived.csv</code>	All years + derived	CHR imputed counts + ratio flags

June 10, 2026

Mental health: data explainer (internal use only)

How to load the latest panel in Python:

```
from functions import latest_file_glob
import pandas as pd

merged_dir = "~/Dropbox/Mental/Data/merged"
path = latest_file_glob(merged_dir, "*_full_merged.csv")
df = pd.read_csv(path, low_memory=False)

# Always normalize the key after loading:
from functions import normalize_panel_key
df = normalize_panel_key(df, dropna=True)
```

6. Appendix

Section content

1. Data sources
2. Key files (csv)
3. Script pipeline overview
4. Merging into the panel
5. Review of panel dataset

6. Appendix

June 10, 2026

Mental health: data explainer (internal use only)

Hardcoded paths

Before running any script on a new machine, update the following:

Script	Variable / Line	What to change
packages.py	db_base = os.path.expanduser("~/Dropbox/Mental")	Dropbox root path
script0a, 0b, 0d	repo = "/Users/allegراسaggese/Documents/..."	GitHub repo path
script0b	USDA_NASS_API_KEY env var	Personal API key
script0b	donor_path = ".../2026-02-23_ag_annual_df.csv"	Backfill CSV path
script0d	interim_override = "/Users/allegراسaggese/Dropbox/..."	Local Dropbox copy path
script0f	HUD_API_TOKEN env var	HUD API token

June 10, 2026

Mental health: data explainer (internal use only)

CHR conversions for incidence rates

In the County Health Survey (CHR) data, variables are recorded with different *rates of incidence*. See table below for steps to convert to same incidence rate (per 100,000 people).

Current unit	Variables	Conversion
Proportion (0–1)	All %_*, excessive_drinking, adult_obesity, adult_smoking, unemployment, uninsured, low_birthweight, children_in_poverty, etc.	$\times 100,000$
Per 1,000	infant_mortality, teen_births	$\times 100$
Per 10,000	social_associations	$\times 10$

CHR conversions for incidence rates

Cont'd table:

Current unit	Variables	Conversion
Already per 100,000	violent_crime, homicides, motor_vehicle_crash_deaths, injury_deaths, hiv_prevalence, STIs, premature_age-adjusted_mortality, child_mortality, preventable_hospital_stays, on-road MVC ER visits	no change
Provider ratio (providers/person)	primary_care_physicians, dentists, mental_health_providers, other_primary_care_providers	$\times 100,000$

June 10, 2026

Mental health: data explainer (internal use only)

CHR conversions for incidence rates

Cont'd table:

Current unit	Variables	Conversion
Not a rate — leave as-is	poor_mental_health_days (avg days/month), poor_physical_health_days, income_inequality (Gini), food_environment_index, air_pollution PM ($\mu\text{g}/\text{m}^3$), median_household_income, premature_death (YPLL per 100k — already standardized), poor_or_fair_health	—

Conversions were done for mental health variables to make them **comparable** numerically. Incidence per 100,000 is also how crime data is reported, and therefore this cleaning standardizes the process.

The rate standardization is in `script1c-merge-dataclean.py`, lines ~489–565, in a block called `# Rate Standardization`. Within the merge dataset, we build the incidence data.

CHR conversions for incidence rates: step by step

- **Part 1** crime counts → per 100k: detects all columns ending in `_crime_fips_level_final`, divides by `population_population_full` × 100,000, writes `*_per100k` columns. 13 crime variables exist.
- **Part 2** CHR variables → per 100k: four explicit lists, each with a known multiplier (see previous table):
 1. Provider ratios (raw ratio like 0.0005) × 100,000 → e.g. `primary_care_physicians_per100k`
 2. Per-1,000 vars × 100 → `teen_births_per100k`, `infant_mortality_per100k`
 3. Per 10,000 vars × 10 → `social_associations_per100k`
 4. 27 proportion (0–1) vars × 100,000 → e.g. `adult_obesity_per100k`